Stack Overflow Developer Survey Analysis: 2024 Global Trends

Hugo Apolinário

June 2025

© IBM Corporation. All rights reserved.

Skills Network

IBM

OUTLINE

- Executive Summary
- Introduction
- Methodology
- Programming Language Trends
- Database Trends
- Dashboard Highlights
- Discussion
- Overall Findings and Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY

- JavaScript, SQL, and HTML/CSS remain the most widely used programming languages.
- PostgreSQL and MySQL dominate database usage, with PostgreSQL also leading future demand.
- Developers prefer platforms like AWS and Google Cloud, and frameworks like React and Node.js.
- Most respondents are aged 25-34, with bachelor's degrees being the most common education.
- Insights support strategic decisions on tech investment, hiring, and learning paths.

INTRODUCTION

- **Purpose:** To uncover global developer technology preferences, trends, and demographics using Stack Overflow's 2024 survey data.
- **Audience:** Tech leaders, HR managers, educators, and developers.
- **Value:** Informs talent strategy, product development, and tech adoption.

METHODOLOGY

Data Source: Stack Overflow Annual Developer Survey (2024)

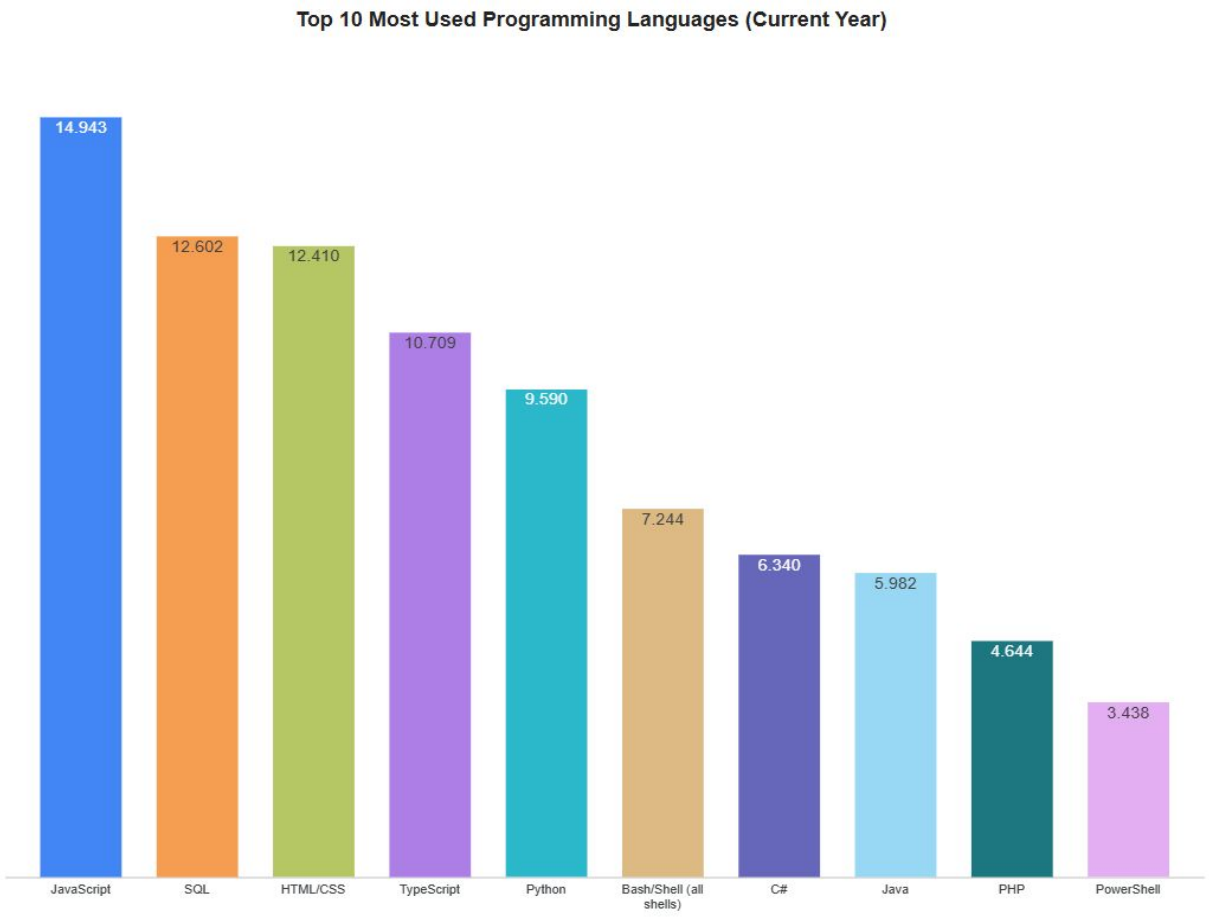
Data Collection: Online global survey: >65,000 responses

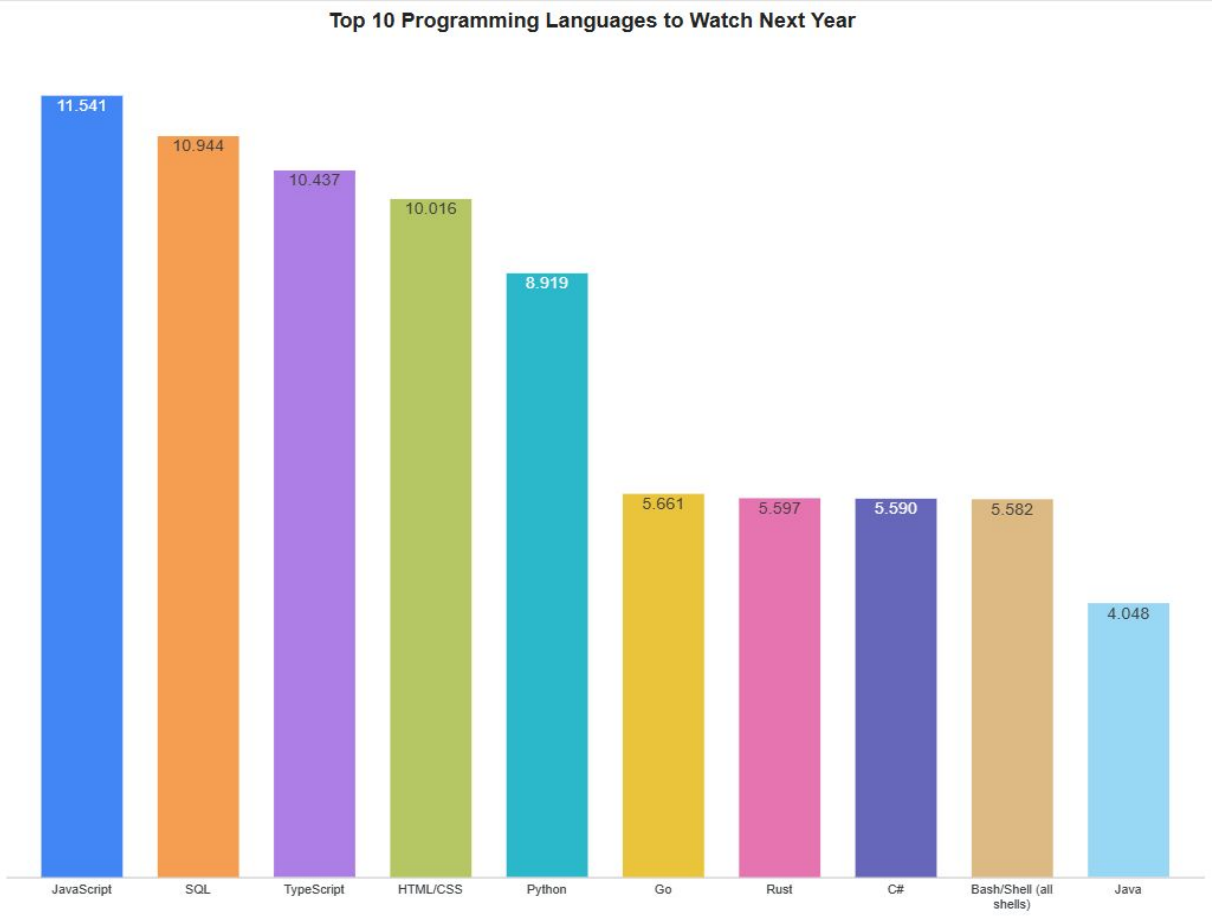
Data Wrangling:

- Removed duplicates, handled missing values
- Normalized multi-response columns (e.g., languages, frameworks)
- Split/aggregated text columns to enable “top 10” analysis

Visualization Tools: Python and Looker Studio

PROGRAMMING LANGUAGE TRENDS





PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

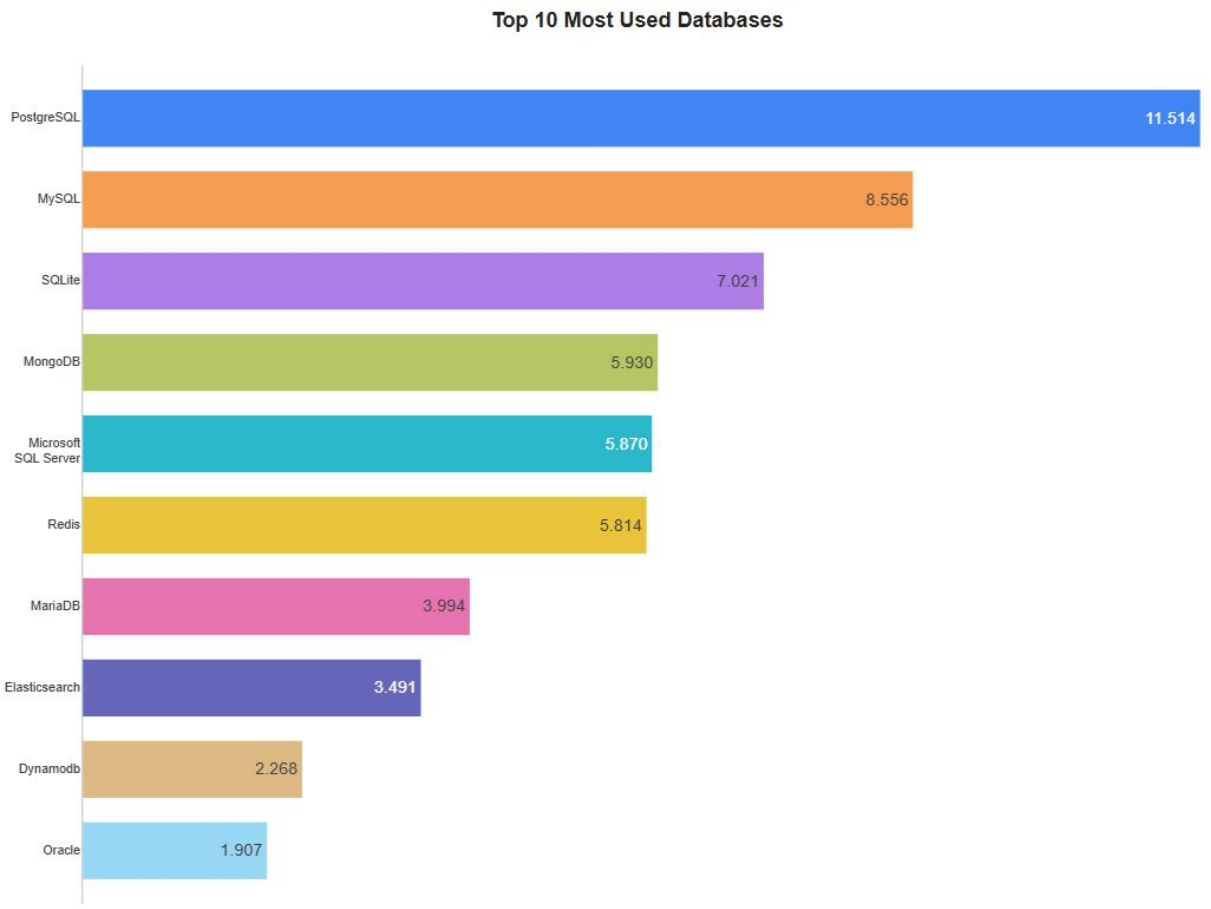
Findings:

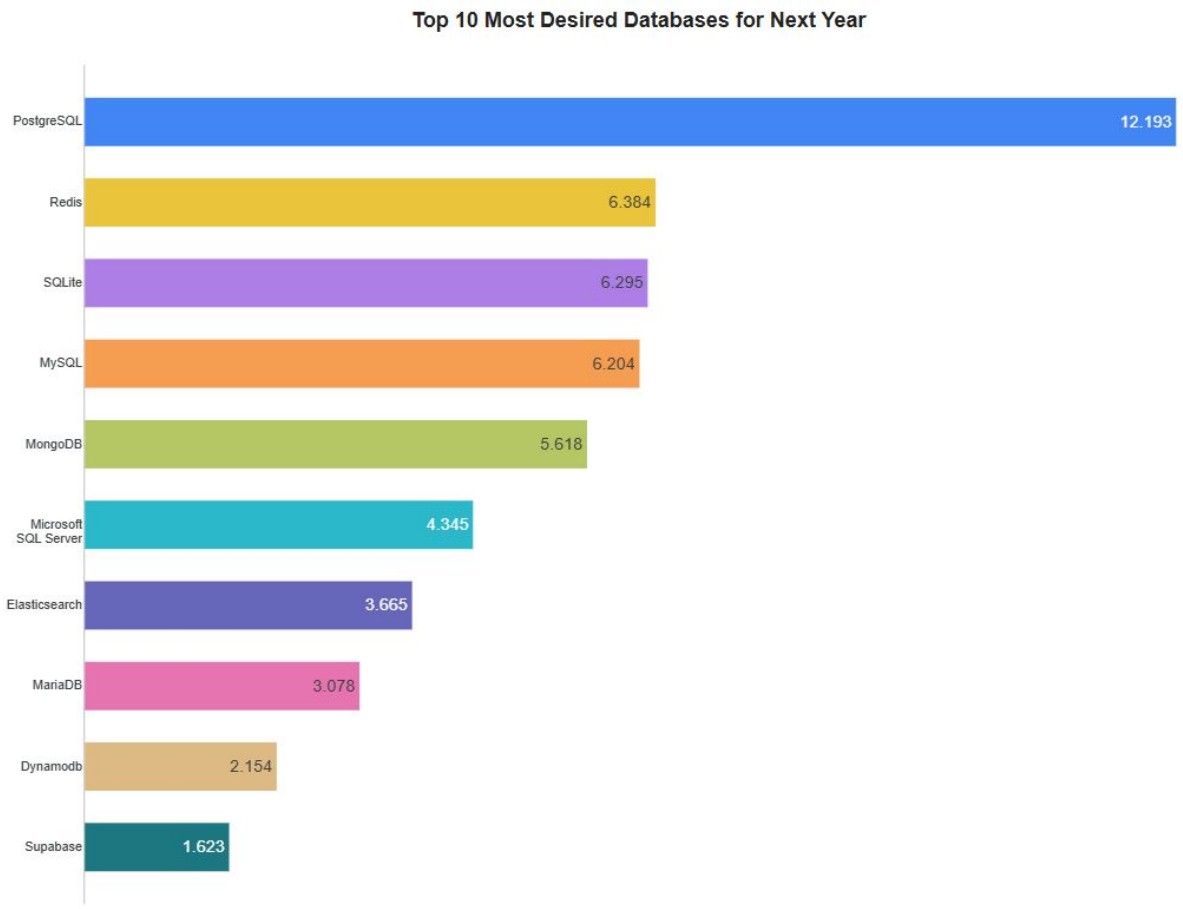
- JavaScript, SQL, and HTML/CSS continue to be the most widely used programming languages.
- Python and TypeScript are experiencing notable growth in usage.
- PowerShell and PHP remain in the top 10 but trail behind in adoption.

Implications

- JavaScript and SQL are foundational, but TypeScript's growth suggests a shift toward safer, scalable JavaScript development.
- Go and Rust stand out as emerging choices for developers planning their next skill upgrade.
- Modern developer demand is trending toward TypeScript and newer, cloud-native languages and frameworks.

DATABASE TRENDS





DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL, MySQL, and SQLite remain the most widely used databases.
- NoSQL databases like MongoDB and Redis have become mainstream choices.

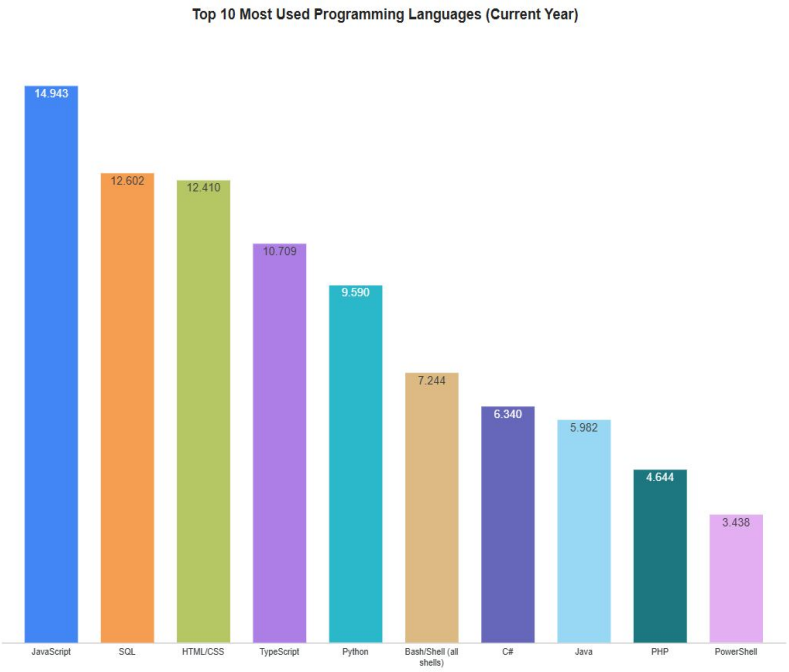
Implications

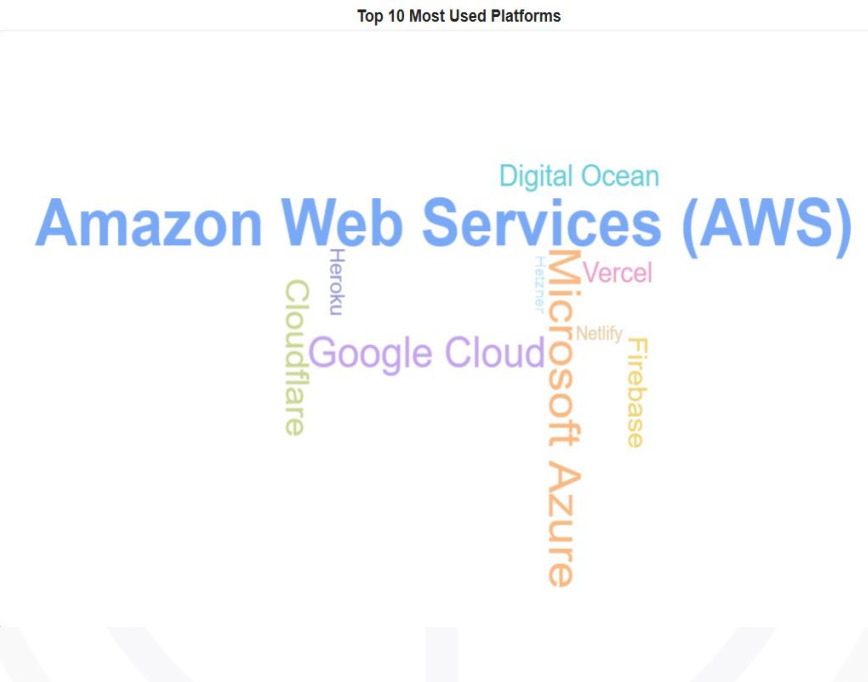
- PostgreSQL is poised to lead future database adoption.
- Redis, MongoDB, and Elasticsearch reflect a shift toward flexible, high-performance data platforms.
- MariaDB and Supabase are rapidly emerging as important new options for developers.

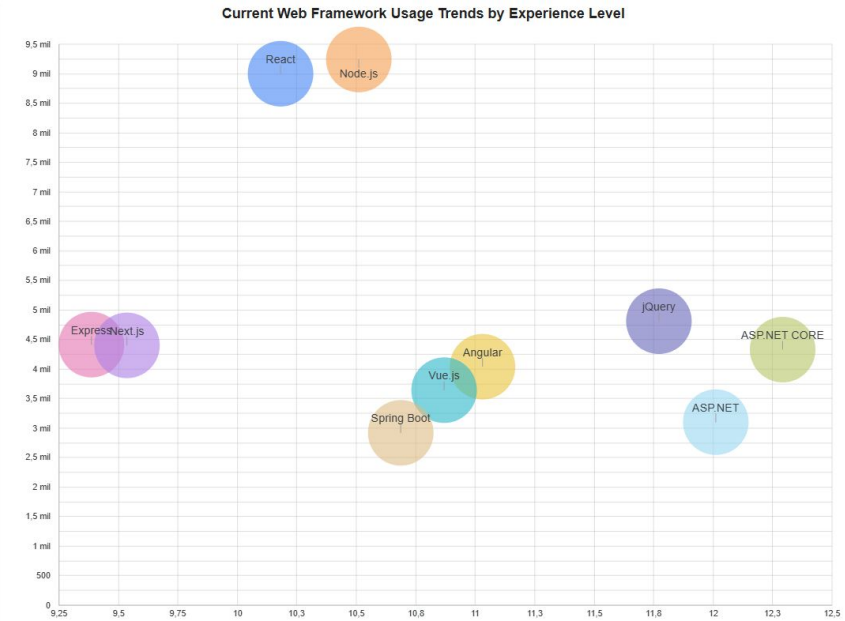
DASHBOARD

The following dashboards provide a quick, visual summary of key survey metrics, allowing for interactive exploration of technology usage, preferences, and respondent demographics.

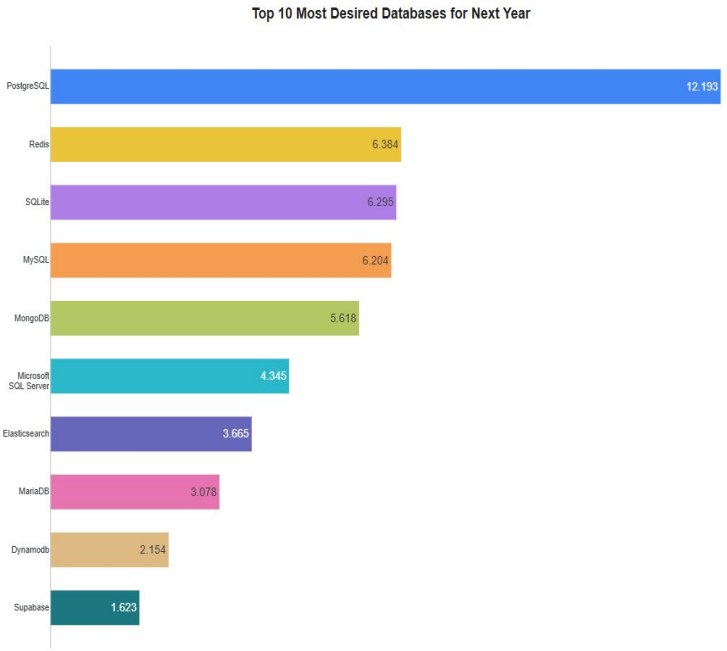
DASHBOARD - Current Technology Usage

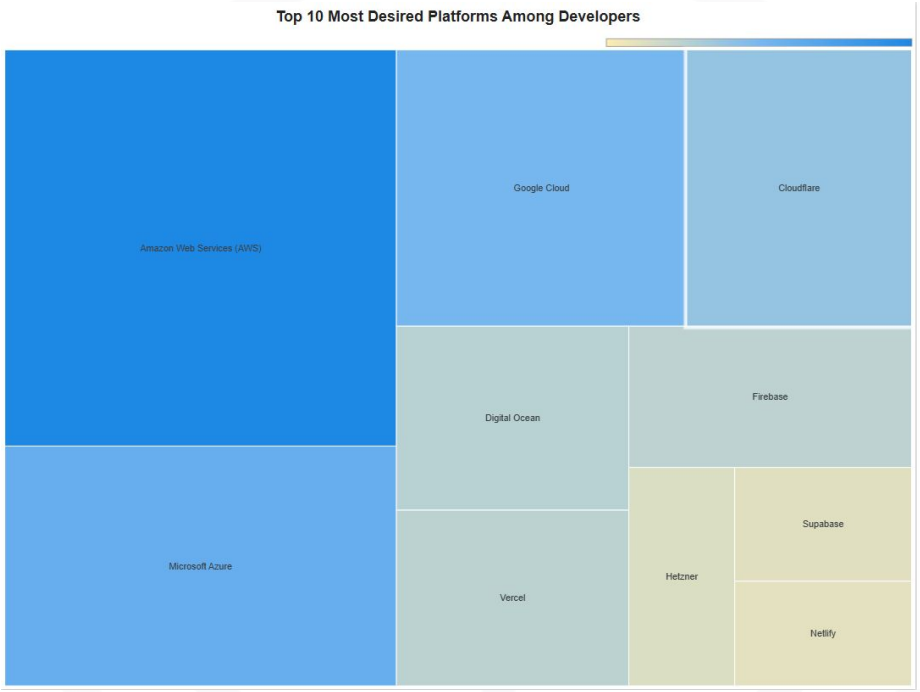


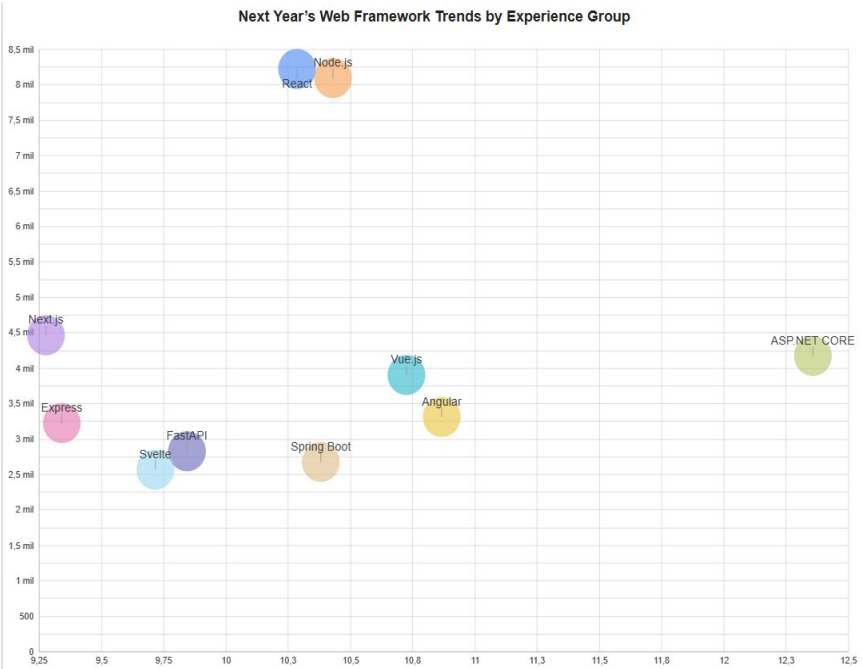




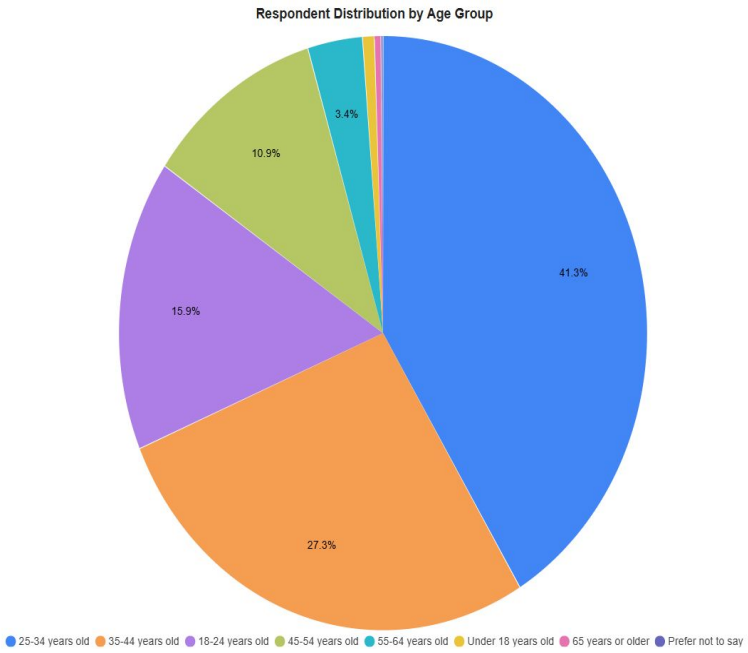
DASHBOARD - Future Technology Trends

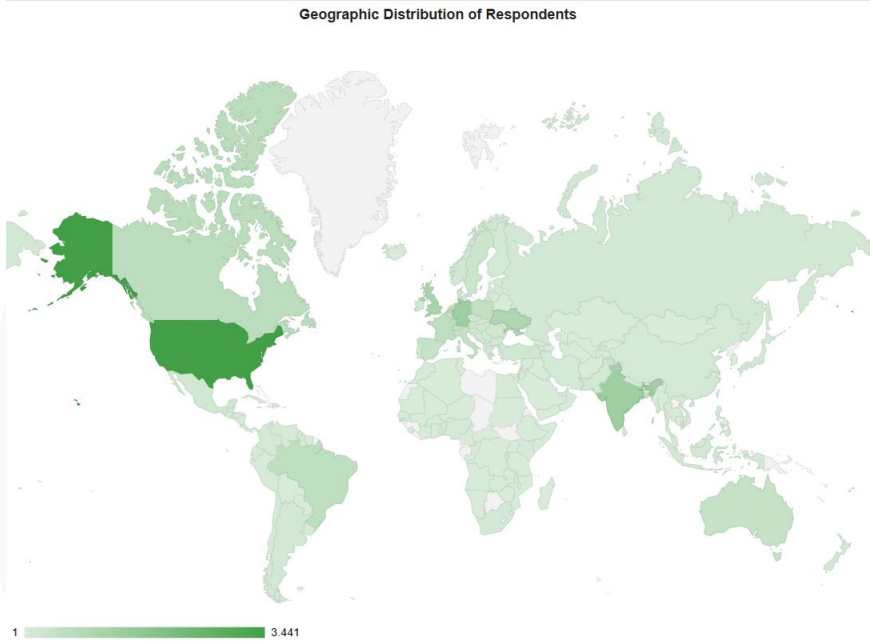


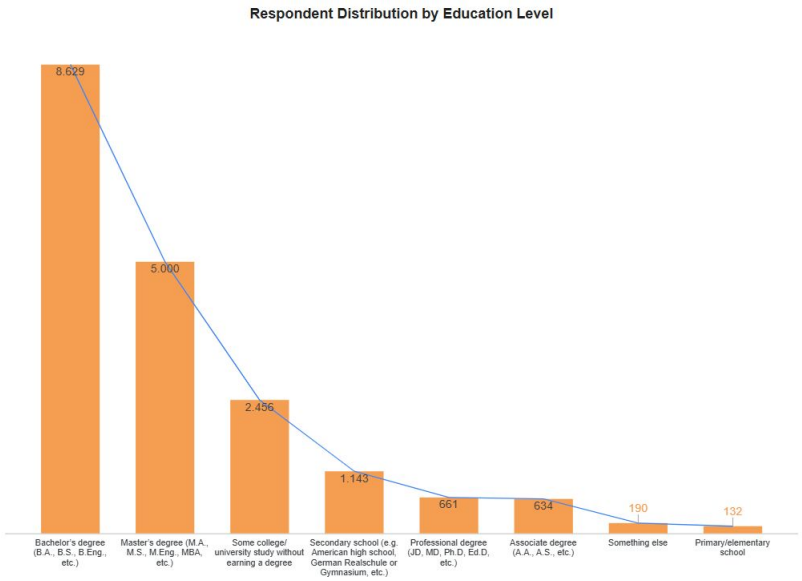




DASHBOARD - Demographics







DISCUSSION

- Programming languages and databases are evolving—TypeScript and Rust are gaining.
- Cloud adoption is nearly universal, with AWS leading.
- Demographics skew young, but there’s meaningful diversity in education and geography.
- Organizations should invest in trending tech and learning resources for their teams.

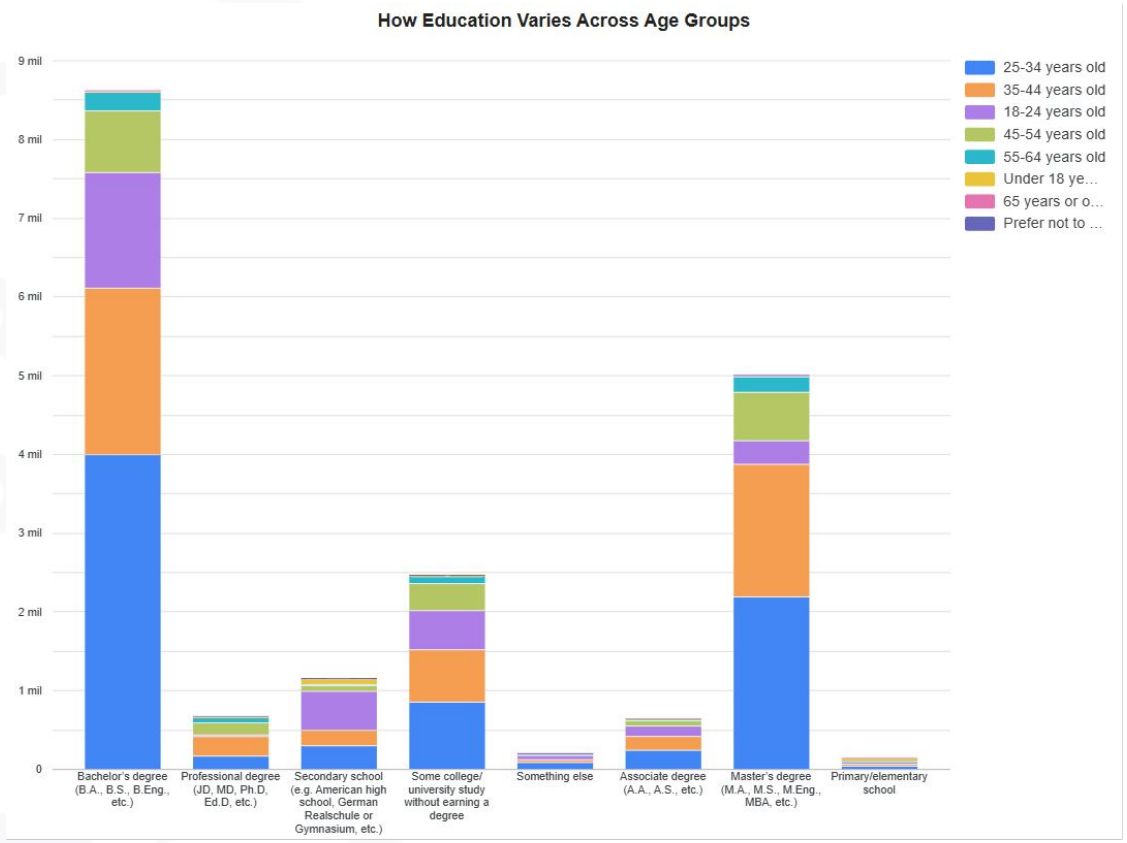
OVERALL FINDINGS & IMPLICATIONS

- **Hiring:** Focus on top languages/platforms for recruitment and upskilling.
- **Technology Investment:** Adopt emerging databases and frameworks to stay competitive.
- **Education:** Promote learning in modern, in-demand languages.
- **Diversity:** Continue to reach and include more countries, age groups, and education backgrounds.

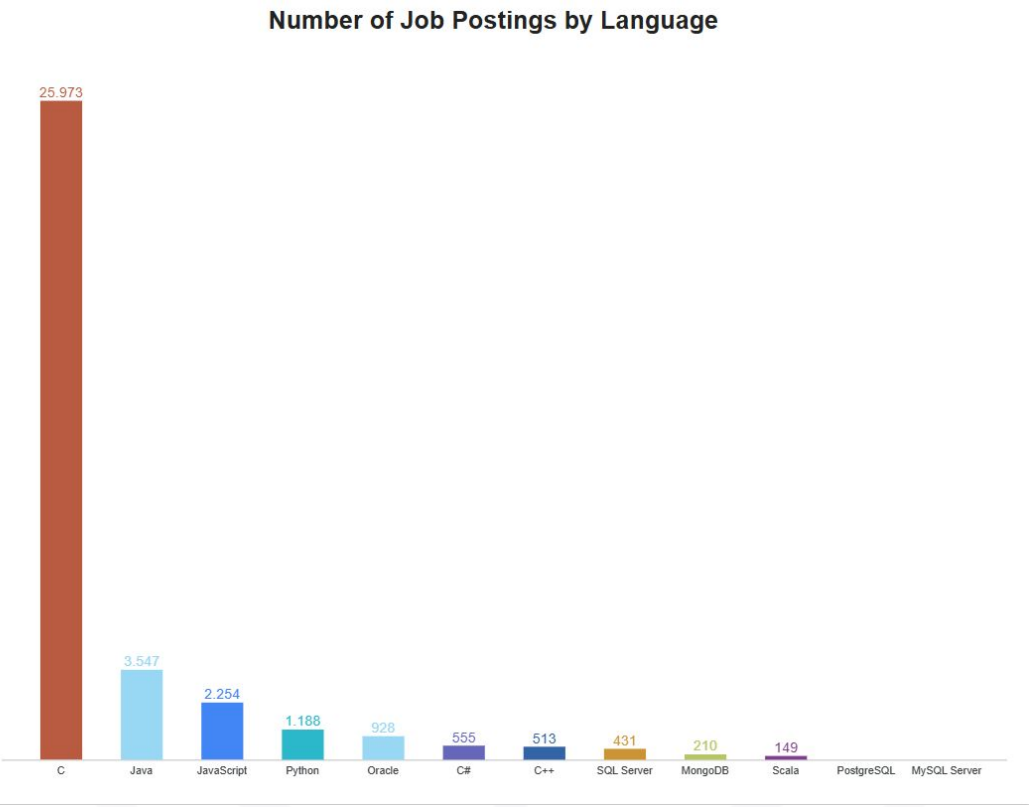
CONCLUSION

- Stack Overflow’s 2024 data provides clear guidance on current and future technology trends.
- Investing in rising languages, cloud platforms, and flexible data tools will prepare organizations for the next wave of innovation.
- Continuous learning and adaptation are key in the evolving developer landscape.

APPENDIX

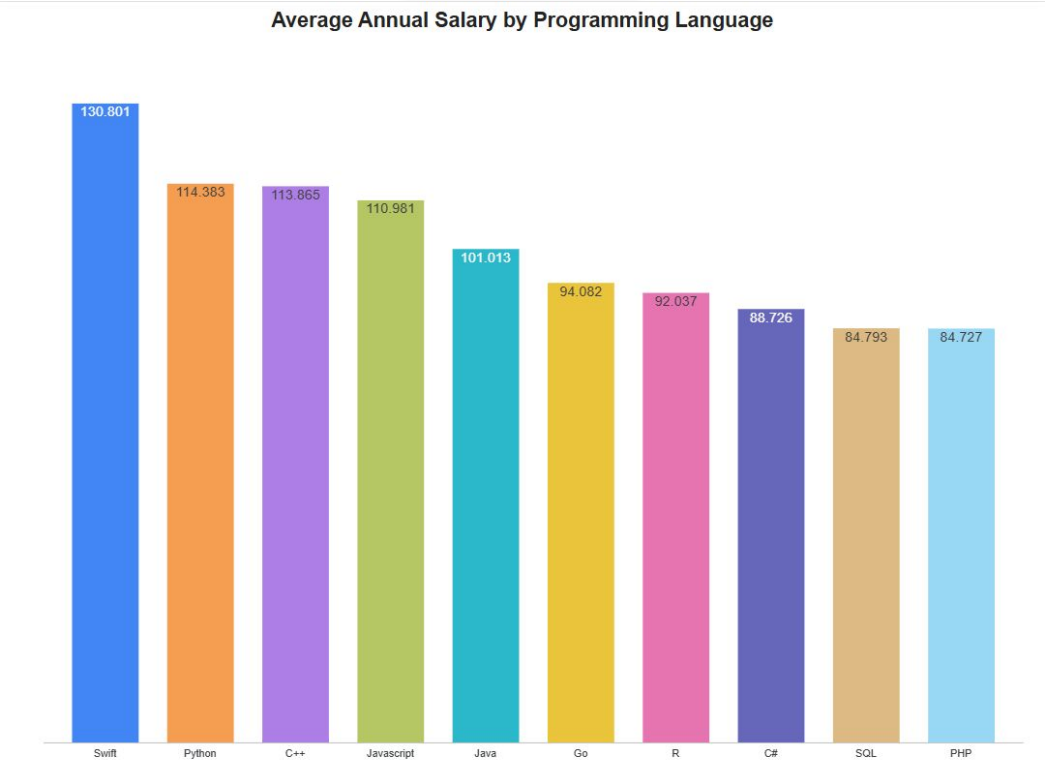

How Education Varies Across Age Groups
9 mil
8 mil
7 mil
6 mil
5 mil
4 mil
3 mil
2 mil
1 mil
0
25-34 years old
35-44 years old
18-24 years old
45-54 years old
55-64 years old
Under 18 ye...
65 years or o...
Prefer not to ...
Bachelor's degree (B.A., B.S., B.Eng., etc.)
Professional degree (JD, MD, Ph.D, Ed.D, etc.)
Secondary school (e.g. American high school, German Realschule or Gymnasium, etc.)
Some college/ university study without earning a degree
Something else
Associate degree (A.A., A.S., etc.)
Master's degree (M.A., M.S., M.Eng., MBA, etc.)
Primary/elementary school

JOB POSTINGS



C dominates job postings in this dataset, followed by Java, JavaScript, and Python. Some technologies like MySQL Server and PostgreSQL show low explicit demand, possibly due to differences in how job requirements are described.

POPULAR LANGUAGES



Swift, Python, and C++ command the highest average annual salaries, while SQL and PHP are at the lower end. The data highlights that specialization in certain languages can significantly boost earning potential.